Carlos Martínez Quintero

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Summary

Mathematics and computer science graduate student with national-level distinctions in mathematical olympiads and ongoing academic interests in algebraic and topological structures, logic, foundations of mathematics, and proof assistants.

Education

University of Oxford, MSc in Mathematics and Foundations of Computer Science
October 2026 - September 2027 (predicted)
Relevant Courses: Machine Learning, Category Theory, Information Theory, Quantum Processes and Computation

Charles University, BSc Computer Science (math-heavy track) **GPA: 1.12 / 4 (best = 1)**
October 2023 - June 2026
Relevant Courses: Algebraic Invariants in Knot Theory, Linear Programming and Combinatorial Optimization, Discrete and Continuous Optimization, Functional Programming, Algebraic Topology 1, Introduction to Combinatorial and Computational Geometry, Set Theory, Probability and Statistics 1 & 2, Formal Mathematics and Proof Assistants, Non-Procedural Programming, Algebra 1 & 2, Propositional and Predicate Logic, Combinatorics and Graph Theory 1 & 2, Automata and Grammars, Algorithms and Data Structures 1 & 2, Computer Systems, Programming 1 & 2, Programming in C++, Linear Algebra 1 & 2, Mathematical Analysis 1, 2 & 3.

Thesis Grade: 1 (Excellent) **State Final Examination Grade:** 1 (Excellent)

Languages spoken

Spanish: Native

English: Native/Bilingual Proficiency (IELTS Academic 8.5/9.0, CEFR C2 - Mastery)

Test scores

IELTS Academic: 8.5/9.0. Taken on December 20, 2025.

Selected fellowships and awards

Atlas Fellowship: Was selected as Atlas fellow. Received scholarship and attended summer program. Awarded May 2022.

Mexican Mathematical Olympiad (OMM) 2021: Gold medal, awarded November 2021.

Mexican Mathematical Olympiad (OMM) 2022: Gold medal, awarded November 2022.

Mexico IMO TSTs: 2022 and 2023, 8th place (best result; top 6 make IMO team)

Selected international mathematics programs

Maths Beyond Limits (2022, 2023), MBL Balkans (2023), Farum Mathcamp (2024), WARP (2022), ESPR (2022) (Given several talks in these).

Mathematical Training and Interests

Algebraic topology, category theory, algebraic structures, logic and foundations, interactions between topology and computation.

Additional training

Intensive problem-solving and theory programs including IMO-style training and advanced international mathematics camps.

Academic Projects

Interactive Tactic-Based Program Synthesis (Bachelor's thesis project): *August 2025-May 2026*

Development of a mathematical formalism to describe type-driven synthesis of functional programs through tactics, alongside a prototype of an interactive system written in Haskell.
Supervisor: Tomáš Petříček, Ph.D.
Grade Awarded: 1 (Excellent)

Teaching

Tutor at Maths Beyond Limits 2024: *September 2024*

Gave a three-day set of lectures on automata theory and formal grammars.

Tutor at MBL Balkans 2025: *April 2025*

Gave a three-day set of lectures on propositional and predicate logic.

Tutor at Maths Beyond Limits 2025: *September 2025*

Gave a three-day set of lectures on theorem proving in Lean, jointly with Ruth Plümer.

Tutor at Maths Beyond Limits 2026: *September 2026*

Gave a three-day set of lectures on category theory.

Work and selected internships

Qualia Research Institute: Independent contractor - Jun 2023 to Aug 2023.

Research internship in theoretical and computational aspects of cognition; literature review and technical research assistance.

Mercor: Math expert - Oct 2024 to Sept 2025, Dec 2025 to Feb 2026

Remote mathematical contractor work involving advanced problem-solving and technical analysis (under NDA).

Programming experience

Proficient:

- **Haskell:** I have written multiple projects in Haskell, and it was also my language of choice for my bachelor's thesis.
- **Lean:** Took a course on Lean in university (Formal Mathematics and Proof Assistants), gave a joint course on dependent type theory and Lean, and have undertaken small formalization projects.
- **Python:** Great familiarity with Python, having used it for different purposes for the last 5 years, including some projects.

Intermediate: C/C++, C#, Typst, LaTeX
Basic: Prolog